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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 3

CLASS 8

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions × 2 marks = 80 marks
Section II	5 Star Questions × 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Operations · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling · Applied Mathematics · Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

Parts follow the AMO syllabus chapters. Chapter 7 is not part of the Class 8 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)**

Q1. Complete the statement: $\sqrt{190}$ is closest to the whole number ____.

- (A) 13 (B) 14 (C) 12 (D) 15

Q2. Real numbers obey the distributive law. Fill the box so that the statement is true:

$$\sqrt{2} \times (3 + \sqrt{2}) = 3\sqrt{2} + \square$$

- (A) $2\sqrt{2}$ (B) 4 (C) 2 (D) $\sqrt{2}$

Q3. The cube 5^3 can be written as the sum of five consecutive odd numbers: $5^3 = 21 + 23 + 25 + 27 + 29$. In the same way, 6^3 is the sum of six consecutive odd numbers. What is the first of these six numbers?

- (A) 33 (B) 35 (C) 41 (D) 31

Part 2 — Operations & Number Properties (Syllabus Chapter 2)

Q4. A sack holds about 1.5×10^6 grains of wheat. Each grain has a mass of about 4×10^{-2} g. Complete the statement: the wheat in the sack has a mass of about ____.

- (A) 6×10^4 g (B) 6×10^{-12} g
(C) 6×10^3 g (D) 6×10^8 g

Q5. Find the missing number:

$$(\square)^{-3} = \frac{125}{8}$$

- (A) $\frac{5}{2}$ (B) $\frac{2}{5}$ (C) $-\frac{2}{5}$ (D) $\frac{8}{125}$

Q6. If $9^x \times 27 = \frac{1}{3}$, what is the value of x?

- (A) 2 (B) -1 (C) -2 (D) -4

Part 3 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

Q7. Rhea takes ₹4,000 out of her piggy bank and puts it in a savings scheme that pays 7% per year simple interest. How much interest does she earn in the fourth year alone?

- (A) ₹840 (B) ₹4,280 (C) ₹1,120 (D) ₹280

Q8. Manav borrows ₹15,000 for 2 years at simple interest to repair his fishing boat in the harbour. The bank raises the rate from 8% to 9% per year before he signs. How much more interest will he pay over the 2 years?

- (A) ₹300 (B) ₹150 (C) ₹2,400 (D) ₹2,700

Q9. A clay pot is sold with a 10% discount followed by a further 25% discount on the reduced price. Which of these always gives the same final price?

- (A) a 25% discount followed by a further 10% discount
 (B) a single discount of 37.5%
 (C) a 17.5% discount followed by a further 17.5% discount
 (D) a single discount of 35%

Q10. The number of honey bees on a bee farm went up by the same percentage in each of two years, from 25,000 to 36,000. By what percentage did the number go up each year?

- (A) 22% (B) 44% (C) 120% (D) 20%

Q11. At a museum gift shop, Saira buys a book priced ₹1,000. She can use a ₹100-off coupon and a 10% member discount, one after the other, in either order. How much more does she pay if she uses the coupon first?

- (A) ₹100 (B) ₹0 (C) ₹10 (D) ₹90

Part 4 — Algebra (Syllabus Chapter 4)

Q12. The identity below has one term missing. Which term completes it?

$$(5p - \square)^2 = 25p^2 - 30pq + 9q^2$$

- (A) 9q (B) 3q (C) 6q (D) -3q

Q13. A square tile has an area of $(9x^2 + 30x + 25)$ cm². What is its perimeter?

- (A) $(12x + 5)$ cm (B) $(6x + 10)$ cm
 (C) $(12x + 20)$ cm (D) $(3x + 5)$ cm

Q14. The school canteen sold some sandwiches on Monday. On Tuesday it sold 8 more than on Monday. On Wednesday it sold twice as many as on Monday. It sold 68 sandwiches in the three days. How many did it sell on Monday?

- (A) 17 (B) 23 (C) 30 (D) 15

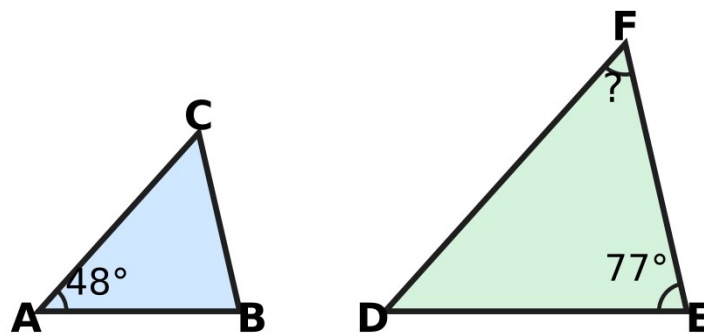
Q15. Both $x^2 - 25$ and $x^2 + 2x - 15$ can be divided exactly by the same expression $(x + k)$. Complete the statement: $k = \underline{\hspace{1cm}}$.

- (A) -3 (B) 5 (C) 3 (D) -5

Q16. On a classroom shelf, $\frac{1}{3}$ of the books are storybooks and $\frac{1}{4}$ are science books. The remaining 30 books are maths books. How many storybooks are there?

- (A) 24 (B) 18 (C) 72 (D) 20

Part 5 — Geometry (Syllabus Chapter 5)

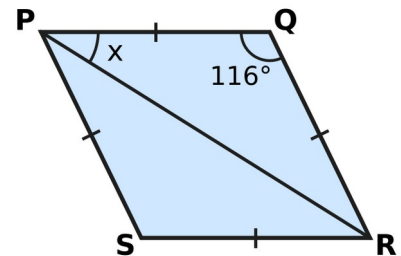


Q17. In the picture, $\triangle ABC \sim \triangle DEF$, with the vertices in this order. $\angle A = 48^\circ$ and $\angle E = 77^\circ$. Complete the statement: $\angle F = \underline{\hspace{1cm}}$.

- (A) 77° (B) 125° (C) 48° (D) 55°

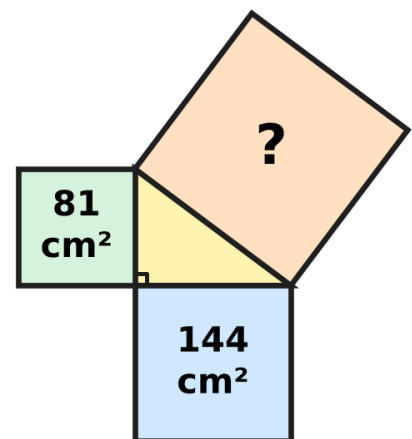
Q18. PQRS is a rhombus, so all four of its sides are equal, as marked in the picture. The diagonal PR is drawn and $\angle PQR = 116^\circ$. What is the angle $x = \angle QPR$?

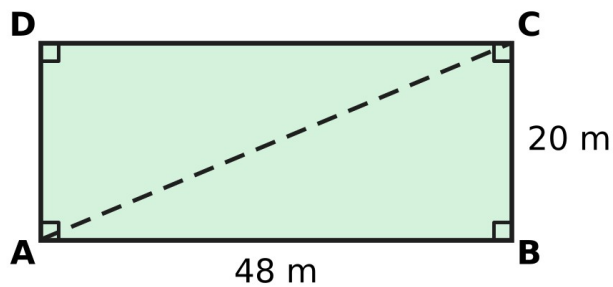
- (A) 64° (B) 32° (C) 36° (D) 58°



Q19. In the picture, a square is drawn on each side of a right-angled triangle. The squares on the two shorter sides have areas 81 cm^2 and 144 cm^2 . What is the area of the square marked "?"

- (A) 225 cm^2 (B) 21 cm^2 (C) 63 cm^2 (D) 15 cm^2



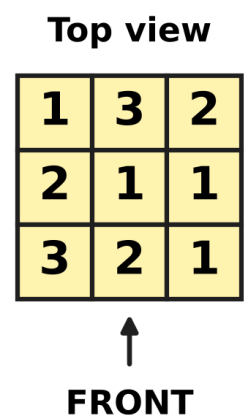


Q20. The picture shows a rectangular playground ABCD, 48 m long and 20 m wide. Nina usually walks from A to B and then to C. Today she walks straight from A to C along the dashed line. How much shorter is her walk today?

- (A) 52 m (B) 68 m (C) 16 m (D) 28 m

Q21. Sahil builds a solid from equal cubes on a 3×3 square base. The picture is the top view; each number tells how many cubes are stacked in that square. He then draws the view from the FRONT, as shown by the arrow. How many squares does the front view have?

- (A) 6 (B) 9 (C) 8 (D) 16



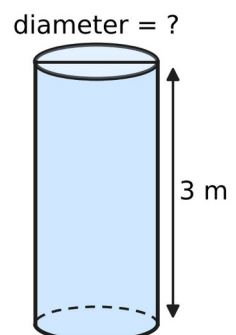
Part 6 — Mensuration (Syllabus Chapter 6)

Q22. A field of area 1.2 hectares is divided into equal square plots of side 20 m, with nothing left over. How many plots are there? (Use 1 hectare = 10,000 m².)

- (A) 300 (B) 30 (C) 3 (D) 600

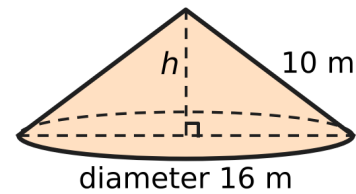
Q23. A cylindrical pillar in the school hall is 3 m tall, as in the picture. Its curved surface has an area of 13.2 m². What is the diameter of the pillar? (Use $\pi = \frac{22}{7}$.)

- (A) 0.7 m (B) 2.8 m (C) 4.4 m (D) 1.4 m



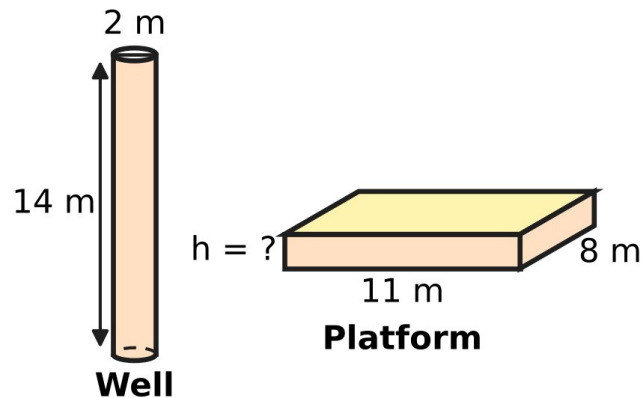
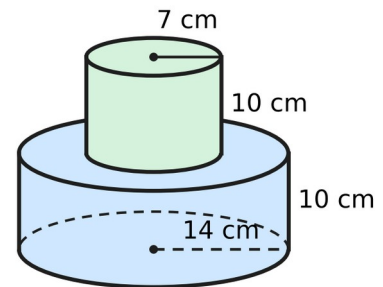
Q24. A conical roof covers a round platform. The base of the roof has a diameter of 16 m and its slant height is 10 m, as in the picture. How high is the top of the roof above the base (h)?

- (A) 6 m (B) 18 m (C) 2 m (D) 8 m



Q25. A solid stand is made of two cylinders, one on top of the other, with their centres in a line. The bottom one has radius 14 cm and the top one has radius 7 cm; each is 10 cm tall, as in the picture. All of it is painted except the face it stands on. What area is painted? (Use $\pi = \frac{22}{7}$.)

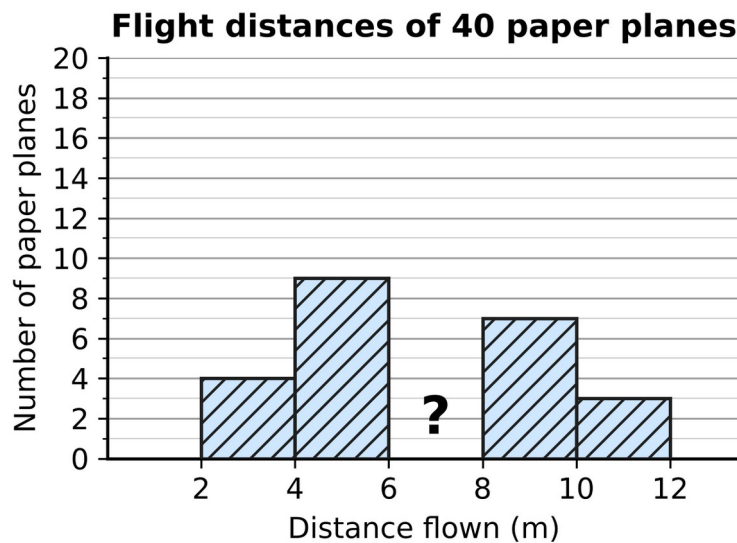
- (A) 2,552 cm² (B) 1,936 cm²
(C) 2,090 cm² (D) 1,474 cm²



Q26. A well is dug in the shape of a cylinder, 2 m in diameter and 14 m deep. All the earth taken out is spread evenly to make a platform 11 m long and 8 m wide, as in the picture. Assume the earth keeps the same volume. How high is the platform, in centimetres? (Use $\pi = \frac{22}{7}$.)

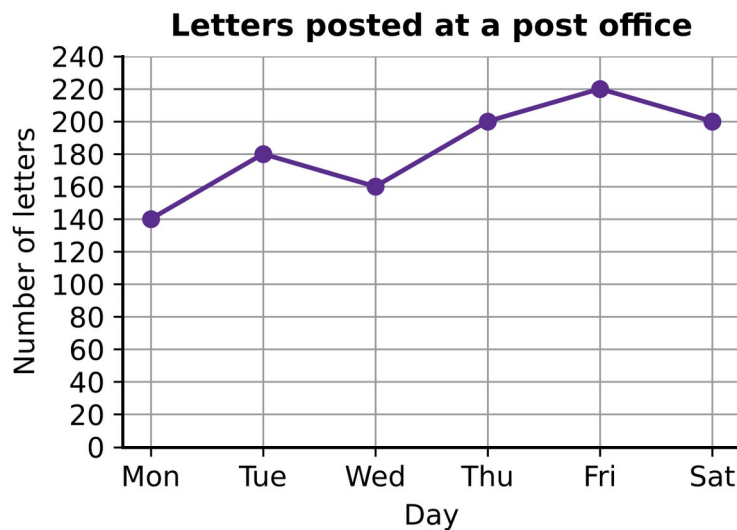
- (A) 200 cm (B) 100 cm (C) 5 cm (D) 50 cm

Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)



Q27. Students flew 40 paper planes and grouped the distances in classes 2-4, 4-6, 6-8, 8-10 and 10-12 m, where 6-8 means 6 m or more but less than 8 m. In the histogram, the bar for the class 6-8 m is missing (marked ?). Complete the statement: the bar for 6-8 m should show ____ planes.

- (A) 17 (B) 23 (C) 9 (D) 40



Q28. The line graph shows the number of letters posted at a post office on each day from Monday to Saturday. On how many of these days were more letters posted than on the day before?

- (A) 5 (B) 4 (C) 3 (D) 2

Q29. In a maths quiz, the 30 students of Section A have a mean score of 14. The 20 students of Section B have a mean score of 19. What is the mean score of all 50 students?

- (A) 16.5 (B) 17 (C) 16 (D) 15

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Q30. Lalit pays ₹5,900 for a board game. This price includes a tax (GST) of 18% on the price before tax. What was the price before tax?

- (A) ₹5,000 (B) ₹4,838 (C) ₹5,882 (D) ₹6,962

Q31. In a clearance sale, Ken buys a bicycle marked ₹6,000 at a discount of 25%. Soon after, he sells it to a friend for ₹6,000. What is his profit percent?

- (A) $133\frac{1}{3}\%$ (B) $33\frac{1}{3}\%$ (C) 75% (D) 25%

Q32. Vinita borrows ₹12,000 for 1 year at 10% per year simple interest to buy a sewing machine. She repays the loan and the interest together in 12 equal monthly payments. How much is each payment?

- (A) ₹1,000 (B) ₹1,200 (C) ₹100 (D) ₹1,100

Q33. Kavin sells school bags at a profit of 20%. Then his supplier lowers the cost price of each bag by 20%, but Kavin keeps his selling price unchanged. What is his new profit percent?

- (A) 20% (B) 50% (C) 40% (D) 44%

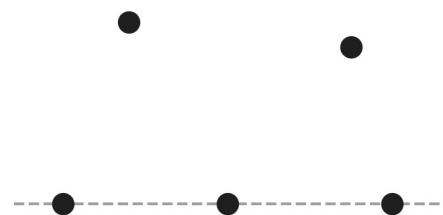
Q34. Payal puts ₹2,000 in a savings scheme at the start of a year. At the start of the next year she puts in another ₹2,000. The scheme pays 10% per year simple interest. How much does she have at the end of the second year?

- (A) ₹4,600 (B) ₹4,400 (C) ₹4,620 (D) ₹4,800

Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)

Q35. Tushar marks five dots on a sheet, as in the picture. Three of the dots lie on one straight line (shown dashed), and no other three dots lie on a straight line. How many different triangles can he draw with all three corners at these dots?

- (A) 10 (B) 3 (C) 9 (D) 6



Q36. Fifteen whole numbers add up to an even number. At least how many of the fifteen numbers must be even?

- (A) 8 (B) 2 (C) 0 (D) 1

Q37. Ella claims: "Every number that is divisible by both 6 and 9 is also divisible by 54." Which of these numbers shows that her claim is false?

- (A) 18 (B) 24 (C) 27 (D) 108

Q38. In the sum $AA + BB + CC = ABC$, each letter stands for a digit. Different letters stand for different digits, and none of A, B, C is 0. (AA is a two-digit number with both digits A, and ABC is a three-digit number.) What is the number ABC?

- (A) 189 (B) 297 (C) 891 (D) 198

Q39. A spinner has three equal parts marked 1, 3 and 7. Harsh spins it three times and adds the three numbers he gets. How many different totals are possible?

- (A) 21 (B) 27 (C) 9 (D) 10

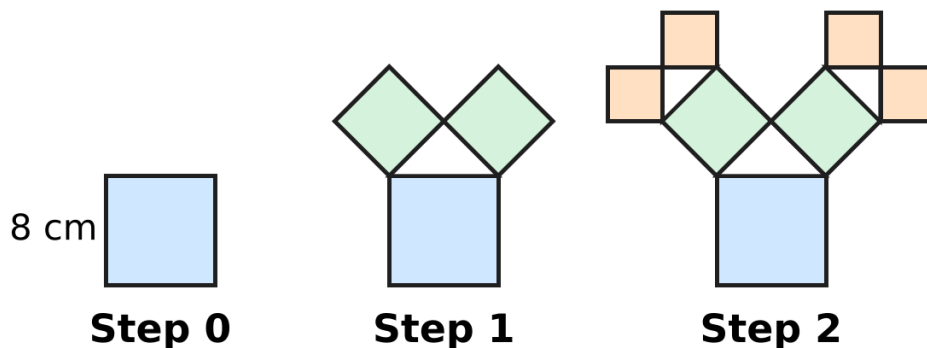
Q40. A number is made by writing the block 48 again and again: 48, 4848, 484848, ... (the block is written k times). Complete the statement: the least value of k for which the number is divisible by 9 is ____.

- (A) 6 (B) 3 (C) 9 (D) 12

SECTION II — Star Questions 5 questions × 4 marks each

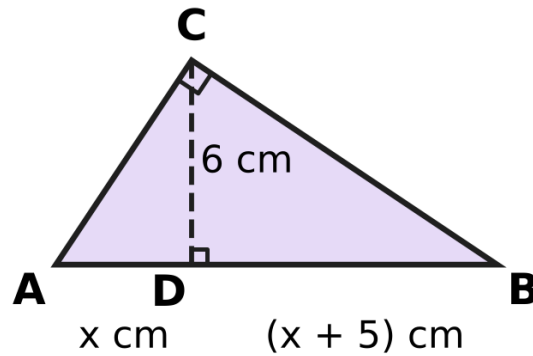
Q41. On a bus, a long-route ticket costs ₹12 and a short-route ticket costs ₹7. On one day, the conductor, Bhavna, collected exactly ₹200 from ticket sales. She sold at least one ticket of each kind, and she sold more long-route tickets than short-route tickets. How many tickets did she sell in all? [4 marks]

- (A) 25 (B) 16 (C) 20 (D) 12



Q42. The picture shows a fractal "tree of squares". Step 0 is a square of side 8 cm. At each step, on the outer side of every square drawn in the step before (the top side in Step 0), a right-angled isosceles triangle is drawn with that side as its longest side. Then a new square is drawn outward on each of the two shorter sides of the triangle. What is the total area of all the squares drawn in Steps 0 to 4, counting each square separately? [4 marks]

- (A) 124 cm^2 (B) 320 cm^2 (C) 64 cm^2 (D) 256 cm^2



Q43. In the picture, triangle ABC has a right angle at C. CD is perpendicular to AB. $AD = x$ cm, $DB = (x + 5)$ cm and $CD = 6$ cm. What is the area of triangle ABC? [4 marks]

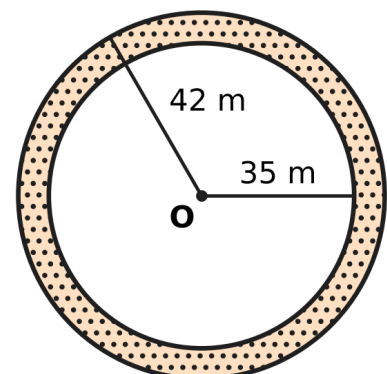
- (A) 39 cm^2 (B) 27 cm^2 (C) 78 cm^2 (D) 36 cm^2

Q44. Vikram grows mangoes and sells them at ₹50 per kg, making a profit of 25% on his cost. Next season his cost per kg goes up by 20%, and he raises his selling price by 8%. What is his new profit percent? [4 marks]

- (A) $11\frac{1}{9}\%$ (B) 13% (C) 20% (D) 12.5%

Q45. A running track is the ring between two circles with the same centre O and radii 35 m and 42 m, as in the picture. A contractor resurfaces the whole track. He charges ₹500 per m^2 but gives a discount of 20% on the total bill. His own cost is ₹300 per m^2 . What is his profit? (Use $\pi = \frac{22}{7}$.) [4 marks]

- (A) ₹15,400 (B) ₹338,800 (C) ₹169,400 (D) ₹554,400



The ring is the track

Answer Sheet AMO Level 1 • Class 8 • Practice Paper 3**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 • Class 8 • Practice Paper 3**Quick answers**

1 B	6 C	11 C	16 A	21 C	26 D	31 B	36 D	41 C
2 C	7 D	12 B	17 D	22 B	27 A	32 D	37 A	42 B
3 D	8 A	13 C	18 B	23 D	28 C	33 B	38 D	43 A
4 A	9 A	14 D	19 A	24 A	29 C	34 A	39 C	44 D
5 B	10 D	15 B	20 C	25 B	30 A	35 C	40 B	45 C

Solutions

Q	Part • level	Answer	Why?
1	Part 1 • Ch1 • Easy	(B) 14	Here $13^2 = 169$ and $14^2 = 196$. The number 190 is 21 more than 169 but only 6 less than 196, so $\sqrt{190}$ is closest to 14.
2	Part 1 • Ch1 • Easy	(C) 2	$\sqrt{2} \times (3 + \sqrt{2}) = \sqrt{2} \times 3 + \sqrt{2} \times \sqrt{2} = 3\sqrt{2} + 2$, because $\sqrt{2} \times \sqrt{2} = 2$.
3	Part 1 • Ch1 • Medium	(D) 31	Here $6^3 = 216$. Six consecutive odd numbers with sum 216 have an average of $216 \div 6 = 36$, so they are 31, 33, 35, 37, 39 and 41. The first is 31.
4	Part 2 • Ch2 • Easy	(A) 6×10^4 g	$(1.5 \times 4) \times 10^6 \times 10^{-2} = 6 \times 10^{6-2} = 6 \times 10^4$ g, which is 60 kg.
5	Part 2 • Ch2 • Easy	(B) $\frac{2}{5}$	$(2/5)^{-3} = (5/2)^3 = 125/8$. So the box holds $2/5$.
6	Part 2 • Ch2 • Medium	(C) -2	Write everything with base 3: $3^{2x} \times 3^3 = 3^{-1}$, so $2x + 3 = -1$, $2x = -4$ and $x = -2$.
7	Part 3 • Ch3 • Easy	(D) ₹280	Simple interest is the same every year: $4,000 \times 7 \div 100 = ₹280$. So the fourth year alone earns ₹280.
8	Part 3 • Ch3 • Easy	(A) ₹300	The rate goes up by 1% a year. Extra interest = $15,000 \times 1 \times 2 \div 100 = ₹300$.
9	Part 3 • Ch3 • Easy	(A) a 25% discount followed by a further 10% discount	Taking 10% off and then 25% off leaves $0.9 \times 0.75 = 0.675$ of the price. Taking 25% off and then 10% off leaves $0.75 \times 0.9 = 0.675$ too, because the order of multiplying does not matter. (A single 35% leaves 0.65, and 17.5% twice leaves about 0.68.)
10	Part 3 • Ch3 • Medium	(D) 20%	Here $36,000 \div 25,000 = 1.44 = 1.2 \times 1.2$. Each year the number was multiplied by 1.2, which is an increase of 20%.
11	Part 3 • Ch3 • Medium	(C) ₹10	Coupon first: $₹1,000 - ₹100 = ₹900$, then 10% off gives ₹810. Discount first: ₹900, then ₹100 off gives ₹800. She pays ₹10 more.
12	Part 4 • Ch4 • Easy	(B) $3q$	$(a - b)^2 = a^2 - 2ab + b^2$. Here $a = 5p$ and $2 \times 5p \times b = 30pq$, so $b = 3q$. Check: $(3q)^2 = 9q^2$.
13	Part 4 • Ch4 • Easy	(C) $(12x + 20)$ cm	Here $9x^2 + 30x + 25 = (3x + 5)^2$, so each side is $(3x + 5)$ cm. Perimeter = $4(3x + 5) = (12x + 20)$ cm.
14	Part 4 • Ch4 • Easy	(D) 15	Let Monday = x . Then $x + (x + 8) + 2x = 68$, so $4x = 60$ and $x = 15$.
15	Part 4 • Ch4 • Medium	(B) 5	$x^2 - 25 = (x - 5)(x + 5)$ and $x^2 + 2x - 15 = (x + 5)(x - 3)$. The common factor is $x + 5$, so $k = 5$.
16	Part 4 • Ch4 • Medium	(A) 24	Let the total be n . The maths books are $1 - 1/3 - 1/4 = 5/12$ of n , so $5n/12 = 30$ and $n = 72$. Storybooks = $72 \div 3 = 24$.

Q	Part · level	Answer	Why?
17	Part 5 · Ch5 · Easy	(D) 55°	Similar triangles have equal corresponding angles: $\angle D = \angle A = 48^\circ$. So $\angle F = 180^\circ - 48^\circ - 77^\circ = 55^\circ$.
18	Part 5 · Ch5 · Easy	(B) 32°	$PQ = QR$, so $\triangle PQR$ is isosceles and $\angle QPR = \angle QRP$. So $x = (180^\circ - 116^\circ) \div 2 = 32^\circ$.
19	Part 5 · Ch5 · Easy	(A) 225 cm^2	By the Baudhāyana–Pythagoras theorem, the square on the hypotenuse equals the sum of the other two squares: $81 + 144 = 225 \text{ cm}^2$.
20	Part 5 · Ch5 · Medium	(C) 16 m	$AC = \sqrt{(48^2 + 20^2)} = \sqrt{2,704} = 52 \text{ m}$. The usual walk is $48 + 20 = 68 \text{ m}$, so today is $68 - 52 = 16 \text{ m}$ shorter.
21	Part 5 · Ch5 · Medium	(C) 8	From the front, each column shows its tallest stack: left column 3, middle column 3, right column 2. The front view has $3 + 3 + 2 = 8$ squares.
22	Part 6 · Ch6 · Easy	(B) 30	The field is 1.2 hectares = $12,000 \text{ m}^2$. Each plot has area $20 \times 20 = 400 \text{ m}^2$. Number of plots = $12,000 \div 400 = 30$.
23	Part 6 · Ch6 · Easy	(D) 1.4 m	Curved surface area = $2\pi rh = 13.2$, so $2 \times 22/7 \times r \times 3 = 13.2$, giving $r = 13.2 \times 7 \div 132 = 0.7 \text{ m}$. The diameter is $2 \times 0.7 = 1.4 \text{ m}$.
24	Part 6 · Ch6 · Easy	(A) 6 m	Radius = $16 \div 2 = 8 \text{ m}$. Height, radius and slant height form a right triangle: $h = \sqrt{(10^2 - 8^2)} = \sqrt{36} = 6 \text{ m}$.
25	Part 6 · Ch6 · Medium	(B) $1,936 \text{ cm}^2$	Curved surfaces: $2 \times 22/7 \times 14 \times 10 = 880$ and $2 \times 22/7 \times 7 \times 10 = 440$. The ring and the small top together make one circle of radius 14: $22/7 \times 196 = 616$. Total = $1,936 \text{ cm}^2$.
26	Part 6 · Ch6 · Medium	(D) 50 cm	Volume of earth = $22/7 \times 1 \times 1 \times 14 = 44 \text{ m}^3$. Height = $44 \div (11 \times 8) = 0.5 \text{ m} = 50 \text{ cm}$.
27	Part 7 · Ch8 · Easy	(A) 17	The four bars shown add up to $4 + 9 + 7 + 3 = 23$. The missing bar is $40 - 23 = 17$.
28	Part 7 · Ch8 · Easy	(C) 3	The line goes up from Monday to Tuesday, from Wednesday to Thursday and from Thursday to Friday. That is 3 days.
29	Part 7 · Ch8 · Medium	(C) 16	Total of scores = $30 \times 14 + 20 \times 19 = 420 + 380 = 800$. Mean = $800 \div 50 = 16$.
30	Part 8 · Ch9 · Easy	(A) ₹5,000	Price before tax $\times 1.18 = 5,900$, so the price before tax = $5,900 \div 1.18 = ₹5,000$. Check: 18% of $5,000 = 900$ and $5,000 + 900 = 5,900$.
31	Part 8 · Ch9 · Easy	(B) $33\frac{1}{3}\%$	Cost price = 75% of ₹6,000 = ₹4,500. Profit = ₹1,500 on ₹4,500 = $1/3 = 33\frac{1}{3}\%$.
32	Part 8 · Ch9 · Easy	(D) ₹1,100	Interest = $12,000 \times 10 \div 100 = ₹1,200$. Total to repay = ₹13,200. Each payment = $13,200 \div 12 = ₹1,100$.
33	Part 8 · Ch9 · Medium	(B) 50%	Let the old cost be ₹100. The selling price is ₹120. The new cost is ₹80, so the profit is ₹40 on ₹80, which is 50%.
34	Part 8 · Ch9 · Medium	(A) ₹4,600	The first ₹2,000 earns interest for 2 years: ₹400. The second earns for 1 year: ₹200. Total = $4,000 + 400 + 200 = ₹4,600$.
35	Part 9 · Ch10 · Easy	(C) 9	Five dots give 10 groups of three dots. The three dots on the line do not make a triangle, so there are $10 - 1 = 9$ triangles.
36	Part 9 · Ch10 · Easy	(D) 1	The sum of an odd count of odd numbers is odd. So the fifteen numbers cannot all be odd; at most 14 can be odd. At least 1 must be even, and 14 odd numbers with one even number do give an even sum.
37	Part 9 · Ch10 · Easy	(A) 18	The number 18 is divisible by 6 and by 9, but not by 54. One such number is enough to show the claim is false.

Q	Part · level	Answer	Why?
38	Part 9 · Ch10 · Medium	(D) 198	$AA + BB + CC = 11(A + B + C)$, which is at most $11 \times 24 = 264$, so the first digit A must be 1 or 2. Trying both cases, only $11 + 99 + 88 = 198$ works: $A = 1, B = 9, C = 8$.
39	Part 9 · Ch10 · Medium	(C) 9	The groups are $1+1+1 = 3, 1+1+3 = 5, 1+3+3 = 7, 1+1+7 = 9, 3+3+3 = 9, 1+3+7 = 11, 3+3+7 = 13, 1+7+7 = 15, 3+7+7 = 17, 7+7+7 = 21$. Since 9 appears twice, there are 9 different totals.
40	Part 9 · Ch10 · Medium	(B) 3	Each block adds $4 + 8 = 12$ to the digit sum, so the digit sum is $12k$. This is a multiple of 9 first when $k = 3$ (digit sum 36). So 484848 is the first one divisible by 9.
41	Section II · Ch4 + Ch10 · Hard	(C) 20	Let a long-route and c short-route tickets be sold: $12a + 7c = 200$, so $200 - 12a$ must be a multiple of 7. Trying $a = 1$ to 16, this happens only for $a = 5$ ($c = 20$) and $a = 12$ ($c = 8$). Since a must be more than c, $a = 12$ and $c = 8$, so she sold 20 tickets.
42	Section II · Ch5 + Ch10 · Hard	(B) 320 cm^2	By the Baudhāyana–Pythagoras theorem, the two squares on the shorter sides together have the same area as the square on the longest side. So each step adds squares with total area 64 cm^2 . Steps 0 to 4 give $5 \times 64 = 320 \text{ cm}^2$.
43	Section II · Ch4 + Ch5 · Hard	(A) 39 cm^2	Triangles ACD and CBD are similar (each has a right angle, and $\angle A = \angle BCD$), so $AD/CD = CD/DB$, giving $x(x + 5) = 36$. Since $4 \times 9 = 36$, $x = 4$ and $AB = 4 + 9 = 13 \text{ cm}$. Area = $\frac{1}{2} \times 13 \times 6 = 39 \text{ cm}^2$.
44	Section II · Ch3 + Ch9 · Hard	(D) 12.5%	Old cost = $50 \div 1.25 = ₹40$ per kg. New cost = $40 \times 1.2 = ₹48$. New price = $50 \times 1.08 = ₹54$. Profit = ₹6 on ₹48 = 12.5%.
45	Section II · Ch6 + Ch9 · Hard	(C) ₹169,400	Area of ring = $22/7 \times (42^2 - 35^2) = 22/7 \times 539 = 1,694 \text{ m}^2$. After the discount he charges ₹400 per m^2 , so his profit is ₹100 per m^2 . Profit = $1,694 \times 100 = ₹169,400$.